

4.1. Реализовать методы Эйлера, Рунге-Кутты и Адамса 4-го порядка в виде программ, задавая в качестве входных данных шаг сетки  $h$ . С использованием разработанного программного обеспечения решить задачу Коши для ОДУ 2-го порядка на указанном отрезке. Оценить погрешность численного решения с использованием метода Рунге – Ромберга и путем сравнения с точным решением.

| №№ | Задача Коши                                                                                                              | Точное решение                                         |
|----|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1  | $y'' + y - \sin 3x = 0,$<br>$y(0) = 1,$<br>$y'(0) = 1,$<br>$x \in [0,1], h = 0.1$                                        | $y = \cos x + \frac{11}{8} \sin x - \frac{\sin 3x}{8}$ |
| 2  | $y'' + y - 2 \cos x = 0,$<br>$y(0) = 1,$<br>$y'(0) = 0,$<br>$x \in [0,1], h = 0.1$                                       | $y = x \sin x + \cos x$                                |
| 3  | $y'' - 2y - 4x^2 e^{x^2} = 0,$<br>$y(0) = 3,$<br>$y'(0) = 0,$<br>$x \in [0,1], h = 0.1$                                  | $y = e^{x^2} + e^{x\sqrt{2}} + e^{-x\sqrt{2}}$         |
| 4  | $x^2 y'' - x(x^2 - 1)y' - (x^2 + 1)y = 0,$<br>$y(1) = 1 + e^{1/2},$<br>$y'(1) = 2e^{1/2} - 1,$<br>$x \in [1,2], h = 0.1$ | $y = \frac{1}{x}(1 + e^{x^2/2})$                       |
| 5  | $y'' - (1 + 2tg^2 x)y = 0,$<br>$y(0) = 1,$<br>$y'(0) = 2,$<br>$x \in [0,1], h = 0.1$                                     | $y = \frac{1}{\cos x} + \sin x + \frac{x}{\cos x}$     |
| 6  | $y'' + 4xy' + (4x^2 + 2)y = 0,$<br>$y(0) = 1,$<br>$y'(0) = 1,$<br>$x \in [0,1], h = 0.1$                                 | $y = (1 + x)e^{-x^2}$                                  |
| 7  | $y'' - 4xy' + (4x^2 - 2)y = 0,$<br>$y(0) = 1,$<br>$y'(0) = 1,$<br>$x \in [0,1], h = 0.1$                                 | $y = (1 + x)e^{x^2}$                                   |
| 8  | $y'' - 4xy' + (4x^2 - 3)y - e^{x^2} = 0,$<br>$y(0) = 1,$<br>$y'(0) = 0,$<br>$x \in [0,1], h = 0.1$                       | $y = (e^x + e^{-x} - 1)e^{x^2}$                        |

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| 9  | $y'' - \left(\frac{1}{x^{1/2}}\right)y' + \left(\frac{1}{4x^2}\right)(x + x^{1/2} - 8)y = 0$<br>$y(1) = 2e,$<br>$y'(1) = 2e,$<br>$x \in [1, 2], h = 0.1$ | $y = \left(x^2 + \frac{1}{x}\right)e^{x^{1/2}}$       |
| 10 | $y'' + y'tg(x) + y \cos^2 x = 0,$<br>$y(0) = 0,$<br>$y'(0) = 1,$<br>$x \in [0, 1], h = 0.1$                                                              | $y = \cos(\sin x) + \sin(\cos x)$                     |
| 11 | $y'' + y'tgx - y \cos^2 x = 0,$<br>$y(0) = 2,$<br>$y'(0) = 0,$<br>$x \in [0, 1], h = 0.1$                                                                | $y = e^{\sin x} + e^{-\sin x}$                        |
| 12 | $(x^2 + 1)y'' - 2xy' + 2y = 0,$<br>$y(0) = 1,$<br>$y'(0) = 1,$<br>$x \in [0, 1], h = 0.1$                                                                | $y = x - x^2 + 1$                                     |
| 13 | $y'' - 2(tgx)y' - 3y = 0,$<br>$y(0) = 1,$<br>$y'(0) = 3,$<br>$x \in [0, 1], h = 0.1$                                                                     | $y = \cos^3 x + \sin x(1 + 2 \cos^2 x)$               |
| 14 | $y'' + 2y'ctgx + 3y = 0,$<br>$y(1) = 1,$<br>$y'(1) = 1,$<br>$x \in [1, 2], h = 0.1$                                                                      | $y = \frac{-0.9783 \cos 2x + 0.4776 \sin 2x}{\sin x}$ |
| 15 | $xy'' + y' = 0,$<br>$y(1) = 1,$<br>$y'(1) = 1,$<br>$x \in [1, 2], h = 0.1$                                                                               | $y = 1 + \ln x $                                      |
| 16 | $(x^2 - 1)y'' - 2xy' + 2y = 0,$<br>$y(2) = 7,$<br>$y'(2) = 5,$<br>$x \in [2, 3], h = 0.1$                                                                | $y = x^2 + x + 1$                                     |
| 17 | $xy'' - (x + 1)y' + y = 0,$<br>$y(1) = 2 + e,$<br>$y'(1) = 1 + e,$<br>$x \in [1, 2], h = 0.1$                                                            | $y = x + 1 + e^x$                                     |

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| 18 | $y'' - \frac{x+1}{x}y' - 2\frac{x-1}{x}y = 0,$ $y(1) = 1,$ $y'(1) = 1,$ $x \in [1, 2], h = 0.1$                           | $y = \frac{e^{2x}}{3e^2} + \frac{(3x+1)e^{-x}}{3e}$                       |
| 19 | $y'' + \frac{1}{x}y' + \frac{2}{x}y = 0,$ $y(1) = 1,$ $y'(1) = 1,$ $x \in [1, 2], h = 0.1$                                | $y = (\cos 2 - \sin 2)\cos(2x^{1/2}) +$ $(\cos 2 + \sin 2)\sin(2x^{1/2})$ |
| 20 | $x(x-1)y'' + \frac{1}{2}y' - \frac{3}{4}y = 0,$ $y(2) = \sqrt{2},$ $y'(2) = \frac{3}{2}\sqrt{2},$ $x \in [2, 3], h = 0.1$ | $y =  x ^{3/2}$                                                           |
| 21 | $x^2y'' - 12y = 0,$ $y(1) = 2,$ $y'(1) = 1,$ $x \in [1, 2], h = 0.1$                                                      | $y = x^4 + x^{-3}$                                                        |
| 22 | $x^2y'' + (x^2 - 2)y = 0,$ $y(1) = 1,$ $y'(1) = 0,$ $x \in [1, 2], h = 0.1$                                               | $y = \sin(x-1) + \frac{1}{x}\cos(x-1)$                                    |
| 23 | $x^2y'' + xy' - y - 3x^2 = 0,$ $y(1) = 3,$ $y'(1) = 2,$ $x \in [1, 2], h = 0.1$                                           | $y = x^2 + x + \frac{1}{x}$                                               |
| 24 | $x^2y'' + (x+1)y' - y = 0,$ $y(1) = 2 + e,$ $y'(1) = 1,$ $x \in [1, 2], h = 0.1$                                          | $y = x + 1 + xe^{1/x}$                                                    |
| 25 | $(x-2)^2y'' - (x-2)y' - 3y = 0,$ $y(3) = 2,$ $y'(3) = 2,$ $x \in [3, 4], h = 0.1$                                         | $y = (x-2)^3 + \frac{1}{x-2}$                                             |

| №№ | Задача Коши                                                                                                                  | Точное решение                                                                |
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| 26 | $x^4 y'' + 2x^3 y' + y = 0,$<br>$y(1) = 1,$<br>$y'(1) = 1,$<br>$x \in [1, 2], h = 0.1$                                       | $y = (\sin 1 + \cos 1) \cos \frac{1}{x} + (\sin 1 - \cos 1) \sin \frac{1}{x}$ |
| 27 | $x^2 y'' - 2xy' + (x^2 + 2)y = 0,$<br>$y(\pi/2) = \pi/2,$<br>$y'(\pi/2) = 1 - \pi/2,$<br>$x \in [\pi/2, \pi/2 + 1], h = 0.1$ | $y = x \cos x + x \sin x$                                                     |
| 28 | $x^2 y'' + 3xy' + 4y - 5x = 0,$<br>$y(1) = 6,$<br>$y'(1) = 8,$<br>$x \in [1, 2], h = 0.1$                                    | $y = 5x + x^2 + x^2 \ln x $                                                   |
| 29 | $x^2 y'' - 3xy' - 5y - x^2 \ln x = 0,$<br>$y(1) = 1,$<br>$y'(1) = 1,$<br>$x \in [1, 2], h = 0.1$                             | $y = \frac{19}{54} x^5 + \frac{35}{54} \frac{1}{x} - \frac{x^2}{9} \ln x $    |
| 30 | $x^2(x+1)y'' - x(2x+1)y' + (2x+1)y = 0$<br>$y(1) = 2,$<br>$y'(1) = 4,$<br>$x \in [1, 2], h = 0.1$                            | $y = x^2 + x + x \ln x$                                                       |

